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Projekt zu Innovation und Wissen - Kreativität und Innovation im regionalen Ökosystem – Erkunden, Vernetzen, Erproben

Carsharing-Projekt in Tanzania

Supervised by:

Prof. Dr. Simone Strambach (Wirtschaftsgeographie)

Prof. Dr. Kathleen Otto (Arbeits- und Organisationspsychologie)

Prof. Dr. Michael Stephan (Innovations- und Technologiemanagement)

Prof. Dr. Gabriele Taentzer (Softwaretechnik)

Vladislav Kargapolov, M. Sc.

Groups and Group Members:

Subgroup “Technical”

Vipul Singh (M.Sc. Computer Science), Sakher Elshteiat (M.Sc. Computer science), Ahmed Elmesery (M.Sc. Computer science), Mohamed Fakhr (M.Sc. Business Informatics), Hikmat Samadov (M. Sc. Data Science)

Subgroup “Marketing / Business Research / Geography”

Tim Pascal Debus (B.Sc. Geography), Nicholas Scheuing (B.Sc. Geography)

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Introduction

The project “Carsharing” has been initiated during the winter semester 2024/2025 by the CIM-Hub, Jörg Brenneisen, Brighton Ammy and Jackson Barnaby. Now, this project is in its second semester. The main goal of this project is to establish an app-supported carsharing service for Tanzanians. According to the stakeholders, the main problem is that due to bad infrastructure, many people do not rely on public transport like trains or buses. Especially students who don’t have the financial means to travel long distances, decide to use alternative methods to reach their goal destination. The stakeholders have noticed many Tanzanians who were sharing cars with strangers as an alternative method, which is self-organized and mostly unstructured. Due to a major lack of security and only sporadic availability, the stakeholders want to provide an app-based solution for safe, reliable and cost-effective car sharing, with a focus on long-distance travel. In addition, the stakeholders want to generate profit of this app, which is planned to be used for supporting Tanzanian students who need financial means.

The app is planned to be a platform, where the user can be the car driver and the passenger. To access the app, the user must create a personal account. In this step, the user must verify himself through an identity confirmation. This account stores information about the user and shows the overall user rating, comparable to rating systems in other services such as Ebay, Uber or Kleinanzeigen. The reviews and ratings are visible to all users of the platform. After an identity confirming process, the user can book a car seat via an app, which checks available drivers on the selected route. The listing for the available route contains information about the driver, the available seats and route duration. To be eligible for a driver, an identity check and a driver’s license check will be mandatory. The driver can then list his route, where other users can join the ride. The driver himself chooses how much he is willing to charge for a seat. On top of these costs, a mandatory fee applies, which finances the insurance and other costs. After users have accepted the listing, the driver must confirm the passenger. The driver can also use the “kick-out” function, which removes the passenger from the reserved seat. After the ride, the driver can also rate the passengers.

As a future milestone, the stakeholders imagine integrating “M-Pesa” as a payment method in their app, a pre-paid and SMS-based money transfer solution. Due to the technical complexity and missing empirical studies around the project, the CIM-team estimates the app to launch earliest in April 2026.

Project progress and organization

Project Progress

At the beginning of the semester, we organized two introductory meetings to review and explain the work completed during the first semester. These sessions focused on revisiting the stakeholders' requirements and presenting the architecture that had been designed for each workflow of the application.

Subsequently, for IT team we conducted two weekly technical meetings to finalize decisions regarding the programming languages, frameworks, and tools for the development phase. Once the technology stack was agreed upon, we divided responsibilities and began working in parallel on different components of the system, including the frontend, backend, and supporting services. Since several of the selected technologies were new to the team, we allocated sufficient time for research and self-learning before starting the implementation. This preparatory phase helped us build the necessary foundation to ensure the quality and maintainability of the system.

In parallel with development, we maintained continuous communication with the stakeholders. Their feedback and updated requirements were carefully documented and considered throughout the development process, ensuring that the implementation remained aligned with both their expectations and the practical realities of the Tanzanian market.

The team in Marburg remained available for support throughout the process in case of any questions or uncertainties. In parallel, initial ideas for data visualization and evaluation methods were developed. These included displaying the results using GIS software (QGIS) as well as visual summaries with charts in Google Forms.

Scientific Background

During this semester, the geography team focused on the implementation phase of the empirical survey developed in the previous term. After finalizing the survey in coordination with stakeholders, we began planning the data collection process. The survey was to be conducted by Jackson Barnaby in Tanzania. For this purpose, clear instructions were prepared, including a target of 200 responses and a deadline of June 29, 2025. The instructions also included distribution methods, documentation requirements, and feedback procedures. The Marburg team remained available for support throughout the process. In preparing for the fieldwork, the team experienced first-hand the organizational and structural disparities

between working conditions in the Global North and South. Delays in communication and different working styles challenged our expectations and planning. At the same time, these experiences contributed to learning processes and required flexibility, cultural sensitivity, and adaptability from all participants.

Empirical Methods / Research Design

The empirical approach remained quantitative, based on a standardized survey. This semester, the focus shifted from survey development to implementation and initial preparations for data analysis. The geography team developed workflows and first ideas for evaluating the results. This included using Google Forms for basic visualizations and QGIS for mapping spatial trends such as frequently travelled intercity routes.

Future students from the geography department could continue to use tools like the ORS Tools Plugin in QGIS to expand the spatial analysis. QGIS holds significant potential not only for interpreting survey data, but also for monitoring spatial patterns during the development of the carsharing service.

Theoretical Foundations

In addition to existing frameworks such as mobility research, the Technology Acceptance Model (TAM) and the concept of digital inequality in the Global South, this semester also brought a stronger focus on transformative competencies. These include the ability to work across disciplines, deal with uncertainty, and act flexibly in dynamic settings. According to RIECKMANN (2012, S.10), such competencies are key for future-oriented education and are particularly relevant in real-world, international project settings.

The process of working with partners under diverse cultural, infrastructural, and organizational conditions directly supported the development of these competencies. This was valuable not only for the project itself, but also for the personal and professional growth of the students involved.

Research Approach

The project continues to follow a deductive-empirical and practice-oriented research design. Theoretical frameworks serve as the basis for formulating hypotheses, which are tested through the ongoing data collection process. This semester added a new applied dimension through planning and partially preparing spatial analyses and GIS-based visualizations. The integration of theory, empirical data, and digital tools like QGIS enhances the interdisciplinary character of the project and improves the accessibility and relevance of the results.

Facilitating Factors

One of the key strengths of our team was its interdisciplinary composition. Team members came from diverse academic backgrounds, including Business Informatics, Computer Science, Geography, and Business Administration. This diversity brings a broad range of perspectives and skill sets to the table.

Each member was able to contribute their unique expertise effectively, which not only enriched the overall project but also allowed for a clear and well-balanced distribution of tasks. As a result, the collaboration was both efficient and productive. In comparison with the first term, the current CIM-team has a larger IT-group, which is very important since the application is in its development phase. Therefore, the recommendation of the first-term team has been integrated into the second-term team.

Lastly, the methodological framework of the project was relatively well defined from the beginning. Although there were still open questions here and there, the overall structure helped us stay on track and follow a logical, goal-oriented approach.

Hindering Factors

Despite the interdisciplinary nature of the team, one of the challenges we faced was the limited level of practical experience among team members. Although we come from relevant fields of study, most of us are (still) in the early stages of our professional development and do not yet have extensive experience in software or app development. As a result, we expect that it will take additional time and effort to reach a stage where a fully functioning application can be developed and implemented. Several uncertainties occasionally led to confusion and required additional clarification and discussion, which slowed down the decision-making process. This factor is a reoccurring issue the CIM-team has faced in the first semester.

During this semester period, the group also faced new challenges. The main problem which hindered the group to work effectively was the lack of internal communication, lack of communication with the stakeholders from Tanzania and with the professors. Due to these issues, the groups were not able to further dive into the development, research for the application and the market strategy behind it. Similar to the situation in the last semester, the response rate from Tanzania still is very low, which made communication very difficult. In the second semester, the geography group unfortunately still does not have enough data to analyze the market situation in Tanzania effectively. Also, the team did not have a direct contact person in Tanzania in this term. Brighton Ammy, who was most of the time communicating

with the CIM-team in the first term, was most of the time not available, which might also have affected the projects efficiency.

In addition, the CIM-team was introduced to a new Tanzanian member who firstly worked under the name of “VATION TECHNOLOGIES LTD” and therefore was considered as a third-party service agent. After communicating to Jörg Brenneisen, he assured that the third party is a Tanzanian student who was supposed to work on the project with the CIM-team. Unfortunately, several meetings with him turned out to be unstructured and inefficient due to communication issues and unclear expectations both from the CIM-team as well as for the third-party. As of now, it is unclear whether he still is part of the project.

Handling of Knowledge / Knowledge Transfer

Knowledge transfer played a central role in our project, as the team members came from different academic backgrounds such as geography, computer science, and business informatics. In order to effectively combine these diverse competencies, a clear line of communication is key for a successful and efficient working. Regular, in-person meetings should be a core value that stands in the group. While communications can also be held online, in-person meetings keep the group members more accountable. During this semester, many members were abroad and therefore not able to meet in person, and consequently, the geography group and the technical group both did not see the urge for discussing the progress amongst each other. These regular meetings are vital to keep all group members informed about the progress of the other subgroups and to share subject-specific insights. After several online meetings, both groups started to work on their own. While it is important that every group member focusses on their specific specialization, regular conversations about the project can spark innovative ideas which then can lead to higher quality outputs.

Also work ethics in Germany and Tanzania vary from each other. The process of knowledge generation, knowledge application and knowledge transfer in Tanzania is in many ways different, which especially stood out to the geography group, since the stakeholders view subjective observations and “visible” problems as sufficient for working on means to solve existing problems, while the geography group prefers to use empirical methods and scientific approaches to analyse and afterwards tackle the problems.

Due to the communication problems within the group and with the stakeholders, the CIM-team has agreed to restructure the group by introducing soft-scale hierarchies. In the past, the group only has been divided by study fields (technical and geography/marketing). This new structure shall boost productivity and

efficiency while making the progress more visible. This is only the first outline of structuring human resources in this project, which is why the current CIM-team recommends revising and if needed update the structure. Additionally, the group plans to develop accountability strategies to measure the progress of the project and to make it easier for new group members to understand how we have worked and tackled issues in the group.

Technical

Planning & Milestones

Before starting the development, we conducted initial research on technology choices to ensure that our solution would be both efficient and sustainable. We compared different frameworks, programming languages, and tools based on criteria such as performance, scalability, community support, and ease of integration. Based on this evaluation, we chose Java with Spring Boot for the backend, starting with a monolithic architecture and planning to refactor to microservices later, and PostgreSQL initially for relational data, migrating to MongoDB Atlas for the microservices setup. We also considered external services like Google Maps API and M-Pesa for future integration. This process allowed us to identify potential trade-offs early on and select technologies that best aligned with our project requirements and long-term goals.

We defined the semester goal as delivering an end-to-end system (backend + temporary frontend) with a clear development roadmap. Our planned milestones included:

- Build a working monolithic backend application (local DB) — foundation for core logic
- Refactor the monolith into microservices (domain split)
- Containerize services
- Provide a temporary frontend prototype to validate backend APIs and visualize workflows
- Integrate M-Pesa payments
- Develop a frontend for the application and link it with the backend

Our development progress was largely successful. We achieved the monolithic backend with complete MVC architecture, successfully refactored into microservices, and implemented full containerization enabling single-command local deployment. The temporary frontend prototype effectively validated our backend endpoints and visualized mapping features. However, both M-Pesa integration and the complete frontend implementation remained incomplete due to time constraints and coordination challenges. The frontend exists as a static design that is yet to be fully linked with the backend.

In parallel to the backend development, the frontend subgroup implemented a demo mobile application using the Flutter framework and Dart programming language. The prototype focused on authentication

workflows, providing login and signup functionality for both normal users and drivers. This demo app was directly connected to Firebase Authentication and Firebase Database, allowing us to test user registration, login flows, and basic data storage in a real environment. Although this prototype currently runs independently from the backend system, it serves as an essential proof-of-concept for validating future integration with the backend APIs developed by the team.

This early prototype also provided stakeholders with a tangible demo of the application's core user experience and allowed us to validate critical authentication workflows before backend integration, reducing future risks and accelerating upcoming development phases.

Technical Approach & Methods Used

Our technical stack centered on Spring Boot with MVC architecture, later refactored using Spring Cloud components for microservices. We implemented Spring Cloud Netflix with Eureka for service discovery, Feign clients for inter-service HTTP calls, and Spring Cloud LoadBalancer for request distribution.

We evolved from local PostgreSQL for the monolith to MongoDB Atlas for microservices, with each service managing its own database for cloud availability and free tier benefits. All services were containerized using Docker, and we selected Google Maps API for mapping functionality based on its accuracy and comprehensive routing features. The project was maintained on GitLab with detailed README documentation and setup instructions.

Obstacles Encountered

Stakeholder communication became challenging when our primary contact's availability decreased and a new representative was introduced, complicating requirement clarifications. The microservices transition required significant learning in distributed systems concepts like Eureka, Feign, and service registration, demanding more implementation effort than expected. M-Pesa payment integration proved complex due to external API dependencies and stakeholder testing requirements, preventing completion within the semester.

On the front-end part, we initially built the frontend in Kotlin (Android-only), which created duplication for iOS, delayed feature parity, and raised maintenance overhead. To streamline delivery, we moved to Flutter,

consolidating to a single codebase for iOS and Android, achieving near-native performance and a fully consistent UI/UX across platforms.

Also, we lacked a production-grade foundation early on. The Kotlin prototypes exposed gaps that slowed iteration on booking flows and real-time features. We then pivoted to Flutter with Firebase (Auth + Firestore/RTDB) as a demo backend, which unblocked maps, tracking, and bookings while we design the long-term architecture.

Solutions & Problem Resolution

We used iterative evaluation for third-party solutions, prototyping and testing components based on accuracy, latency, cost, and developer productivity. The backend lead conducted focused research and built validation prototypes for key approaches like mapping services and microservice patterns. We maintained comprehensive documentation in the repository README, including setup steps, environment variables, and Docker Compose instructions, enabling consistent local development across the team.

On the front end, we leverage Flutter's high-performance rendering and Dart's AOT compilation to deliver smooth, native-like maps, real-time tracking, and responsive booking flows. A single codebase for iOS and Android—enhanced by Hot Reload—accelerates delivery, reduces cost, and ensures a consistent, polished UI/UX as we scale. Combined with a rich plugin ecosystem (maps, geolocation, payments, push notifications, background tasks) and Dart's sound typing, this approach improves reliability, testability, and long-term maintainability for our car-sharing platform.

Supporting Factors / What Went Well

Several factors contributed significantly to our project's success. The architecture artifacts from semester 1 provided a clear and comprehensive roadmap for implementation, eliminating much of the initial design uncertainty. Our technology choices proved well-aligned with project constraints, as Google Maps API and MongoDB Atlas delivered the required accuracy, cost-effectiveness, and ease of use we needed.

The validation process was greatly accelerated through effective tooling. Postman and the temporary frontend prototype enabled rapid backend API validation and thorough testing of our driver-matching logic. Additionally, our focus on comprehensive documentation and maintaining a well-structured GitLab repository with clear README files and setup instructions significantly improved knowledge sharing across the team.

Knowledge Transfer Within the Team

All project code and documentation are centrally hosted in our GitLab repository, with the README containing detailed branch explanations, comprehensive setup steps, and environmental configuration examples. We implemented a fully documented Docker Compose setup alongside standardized .env configuration to minimize friction for teammates when running the full stack locally.

One comprehensive walkthrough session was conducted to present the backend design, explain the microservices architecture, and demonstrate system execution procedures. As the frontend and other components reach completion, additional knowledge transfer sessions can be scheduled to ensure continued team alignment and comprehensive understanding of the evolving system.

Team

- **Vipul Singh** — worked on backend development end-to-end: built the monolithic system, migrated it to microservices, set up databases (PostgreSQL → MongoDB Atlas), cloud integration, Docker containerization, and also created a temporary frontend prototype for testing.
- **Mohamed Fakhr** — I contributed to the project by supporting the database design and schema creation. I also prepared reports such as the system architecture diagram
- **Ahmed Elmesery** — worked on frontend development, implemented a demo mobile application with Flutter/Dart. Focused on authentication screens (login/signup for users and drivers) connected to Firebase Authentication and Database for testing functionality
- **Sakher Elshteiat** — collaborated on frontend development, supporting the Firebase integration and ensuring authentication workflows were validated in the demo app, ensured alignment with backend architecture to support future integration.
- **Hikmat Samadov** – conducted the initial research on technologies, including programming languages, databases, and system architecture (evaluating microservice vs. monolith). He also developed a demo mobile app using Kotlin Multiplatform as an early proof of concept, but the development was discontinued after concluding that KM was not the best fit for the project.

Reflection & outlook

Decisions that proved problematic or incomplete

- The microservice path required substantially more time than originally estimated due to the educational overhead and the amount of supporting infrastructure code required. This learning cost reduced time available for feature work (notably, M-Pesa integration).
- M-Pesa payment integration was not completed this semester; it remains a high-priority technical task and requires more stakeholder coordination and testing with the Tanzanian provider environment.

Summary of results

- Deliverables achieved this semester: a working monolithic backend, migration to microservices, driver-matching and ride allocation logic, Google Maps integration, data hosted on MongoDB Atlas, Containerized services, and a temporary frontend prototype for API validation.
- The system is in a state where other team members can run and test the stack locally and extend features or complete integrations.
- A Flutter-based demo mobile app was created, linked to Firebase Authentication and Database, supporting login and signup screens for both users and drivers. This prototype successfully demonstrated authentication flows, gave stakeholders an interactive preview of the planned system, and laid the groundwork for future frontend–backend integration.

Next steps & recommendations

Priority items

1. Integrate M-Pesa payments — coordinate with stakeholders and allocate time for provider testing.
2. Connect and complete the frontend
3. Deploy to a cloud environment (staging/production) beyond local Docker

4. Security hardening — review authentication/authorization, secure environment variables, and add input validation and logging for sensitive endpoints.
5. Testing & QA — implement comprehensive unit / integration tests and create test cases (including automated integration tests for common user flows).
6. Documentation & onboarding — continue to improve developer docs and add user/deployment guides for future team members

Reflection and Conclusion

1. The methodological basics from the first semesters of the B.Sc. in Geography are essential for effective and project-promoting work. As theoretical foundations, the modules "Methods of Empirical Social Research" (from the methodological competencies), "Economic and Service Geography" and "Geographies of Peripheral Spaces" (from the basic competencies) proved particularly helpful, as they enabled a differentiated approach and direct methodological applicability. In addition, the geography group concluded that soft skills and management skills will have to play a more dominant role in future semesters.
2. During this phase of our master's project, we had to apply a lot of what we learned in Masters in Computer Science to real-world system design. Our knowledge from "Implementation of Database Systems" and "Geo Databases" really came in handy when designing how our system stores and retrieves data using PostgreSQL and Redis. Courses like "Big Data Systems" helped us think about scalability and making sure our services could handle growing demands. Overall, communication between stakeholders in Tanzania and collaboration with lecturers should be given higher priority in the following semesters to better define goals and eliminate any misunderstandings between stakeholders, lecturers, and project participants. The interim discussions with lecturers have always proven to be particularly helpful and have increased the goal orientation of the group members, especially in the geography team. The problem definition and project goal should also be clear and unambiguously recorded on paper by the stakeholder at the beginning of a project, serving as a basis for the project and eliminating misunderstandings (such as the monetization of the project for the benefit of the stakeholders).

Literature

RIECKMANN, M. (2012): Future-oriented higher education. Which key competencies should be fostered through university teaching and learning. Futures 44/2: 127–135.